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NPS/003/024 – Technical Specification for 66kV & 132kV Gas Insulated Switchgear

1. Purpose

This document provides the technical specification for 66kV and 132kV gas insulated switchgear (GIS), and supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
No existing GIS Spec		
No existing GIS Spec		

2. Scope

This specification covers the technical requirements for 66kV and 132kV indoor and outdoor GIS for use on the NEDL and YEDL distribution networks. It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum to Supplier Requirements.

3. Technical Requirements

3.1 General Requirements

Switchgear and switchgear components shall comply fully with the requirements of the latest version of Electricity Network Association Technical Specification (ENATS) 41-37 Parts 1-4 and the latest versions of the standards and regulations to which ENA TS 41-37 refers.

Switchgear shall preferably have been assessed by the ENA Switchgear Assessment Panel (SAP) and have been awarded an ENA Notice of Conformity. This shall be confirmed by the supplier.

Appendix 1 of this document details the requirements where options are given within ENATS 41-37, where the requirements of ENATS 41-37 have been supplemented, or where it is felt necessary to confirm that requirement. Note particularly the specified fault-break duties and X/R ratios.

Appendix 2 contains a pro-forma in which the technical performance of the switchgear shall be declared.

Appendix 3 contains an Addendum to Supplier Requirements which lists project specific requirements.

3.2 Application

The switchgear shall be either indoor or outdoor as specified in Appendix 3

3.3 Rated Voltage

The rated voltage shall be either 72.5kV or 145kV as specified in Appendix 3

3.4 Insulation Level

The rated insulation levels shall be as specified in Appendix 1 corresponding to the specified rated voltage.

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3.5 Air insulated bushings

Where high voltage connections are through external bushings, creepage distance and pollution class shall be as stated in Appendix 1.

Personal safety distances to any exposed HV conductors shall be as specified in Appendix 1.

3.6 Normal rated current: 66kV supply point substations

Incomers and bus sections shall be rated at 2000A. Feeders shall be rated at 1250A

3.7 Normal rated current: 66kV primary substations

Incomers, bus sections and feeders shall be rated at 1250A

3.8 Normal rated current: 132kV supply point substations

Incomers, bus sections and feeders shall be rated at 2000A

3.9 Rated peak and short time withstand currents

Shall be as specified in Appendix 1.

3.10 Rated short circuit breaking current

66kV circuit breakers shall have a rated short circuit breaking current of 31.5kA at an X/R of 14.

132kV circuit breakers shall have a rated short circuit breaking current of 40kA at an X/R of 14.

3.11 Transient recovery voltages

Transient recovery voltages related to the rated circuit breaking currents shall be as specified in ENATS 41-37.

3.12 Circuit breaker rated operating sequence

The rated operating sequence shall be 0 – 0.3s – CO – 3min – CO

3.13 Circuit breaker rated capacitance switching current

The circuit breaker shall be rated Class 2 as defined in ENATS 41-37

3.14 Mechanical endurance

Switchgear shall be rated Class M1.

3.15 Auxiliary and Control Equipment

Auxiliary and control equipment shall generally comply with section 5.4 of ENATS 41-37

Auxiliary voltages shall be 110V dc

CT secondary terminals shall be of the spring loaded or cage clamp design, or spring washer on stud design.

Secondary terminals for CT buswiring shall have a shorting/disconnecting facility.

Secondary terminals for SCADA wiring shall be of the knife disconnecting type and be equipped with 2.3mm test sockets.

3.16 Current Transformers

Current transformers shall be in accordance with BS EN 60044-1, as expanded or modified in Appendices 1 and 3 of this document.

3.17 Test bushings

Test bushings shall be suitable for AC power frequency or very low frequency (VLF) at the voltages stated in Appendix 1

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Test bushings provided for pressure testing switchgear shall be suitable for the voltages that are stated in Appendix 1 of this document.

Test bushings shall be provided to allow the primary injection of all CTs without requiring internal access to any gas compartment.

3.18 Key interlocks

Key interlocks shall be provided to comply with the substation interlock scheme.

3.19 Outage constraints

The equipment shall be capable of being configured such that installation, extension, operation and maintenance procedures may be carried out with a maximum of one adjacent circuit and one section of busbar out of service simultaneously.

3.20 Earth Switches

Provision shall be made to apply earths to busbars, disconnectors, circuit breakers and power cables to allow maintenance procedures to be undertaken.

Earth switches used to discharge feeders and busbars shall be of class E1 as defined in IEC 62271-102

Earth switches used for earthing feeder circuits shall be rated Class B for induced current switching.

The positions on the switchgear at which additional earths may be fitted shall be clearly identified.

3.21 Disconnectors

Busbar disconnectors on double busbar switchboards shall be rated for bus transfer duty and be suitably interlocked. Disconnectors shall have the same mechanical endurance as the circuit breaker.

3.22 SF6 Gas connections

The gas service connection for each gas compartment shall be readily accessible without the use of special access equipment.

Filling points shall be fitted with self sealing valves.

3.23 Pressure/Density Indication

Further to the requirement of ENATS 41-37 the system shall be capable of detecting a gas loss equivalent to 0.5% of the gas compartment volume per annum or shall provide trend analysis facilities.

3.24 SF6 Gas Alarm Scheme

The grouping of "low alarms" at the normal substation control point shall be such that the section of the substation that requires isolation from the system can be readily identified.

Disconnectors shall be provided with lockout facilities to prevent operation at less than the minimum functional pressure or density.

3.25 Condition Monitoring

On line condition monitoring (including partial discharge monitoring) should be offered as an option.

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3.26 Pressure Relief

Pressure relief devices shall be installed so as to eject debris away from normally accessible areas. They shall be set to minimise danger to personnel.

3.27 Identification of Gas Compartment Partitions

The position of each gas compartment partition shall be clearly identified.

Labels shall be fixed to the enclosure at each gas compartment partition showing the identifier of the gas compartment at each side of that position.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 41-37 Parts 1-4	Switchgear for use on 66kV to 132kV Distribution Systems
BS 381C: 1996	Specification for colours for identification, coding and special purposes

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	19/06/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	19/06/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D.Blackledge	Standards Engineer	D.Blackledge	22/06/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	28/06/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	21/06/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	21/06/07

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APPENDIX-1: Technical Specification**66kV Switchgear**

Summary of Technical Requirements and Ratings	ENA TS 41-37	UNIT	
System Particulars			
Voltage		kV	66
Frequency		Hz	50
Number of Phases			3
Neutral Earthing			Solid
General Characteristics	Part 1 Clause		
Number of poles			3
Class – indoor/outdoor	2.1.1		Indoor / Outdoor
Rated voltage	4.1	kV	72.5
Rated insulation level. Lightning impulse level	4.2	kV	325 375 (across isolating gap)
Power frequency withstand			140 160 (across isolator gap)
Rated frequency	4.3	Hz	50
Rated normal current	4.4	A	<u>Supply point substation:</u> Incomer, bus-section and busbar: 2000A Feeder: 1250A <u>Primary substation:</u> Incomer, section, busbar and feeders:1250A
Rated short-time withstand current	4.5	kA	31.5
Rated peak withstand current	4.6	kA	78
Rated duration of short circuit	4.7	s	3
Rated supply voltage of opening and closing devices and auxiliary and control circuits: Closing and tripping Indication Control	4.8	V	110 110 110
Rated supply frequency of closing and auxiliary circuits.	4.9	Hz	DC
	Part 2 Clause		

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Cable testing	5.202		AC tests either VLF or power frequency will need to be applied at 58kV for 15 minutes.
Circuit breaker characteristics	Part 3 Clause		
Rated short-circuit breaking current. Equal to rated short-time withstand current.	4.101	kA	31.5 @ X/R = 14
Transient recovery voltage related to the rated short-circuit breaking current.	4.102		In compliance with ENA Technical Specification 41-37
Rated short-circuit making current	4.103	kA	In compliance with ENA Technical Specification 41-37
Rated operating sequence.	4.104		0 – 0.3 s – CO – 3min - CO
Rated capacitive switching currents : Rated cable-charging current.	4.107	A	Class 2
Rated DC time constant.	4.109	ms	Manufacturer to state (see specified fault breaking current)
Classification of mechanical operations M1 or M2	4.110		Class M1
Additional key interlocks			Provision for 1 additional key interlock.
CT secondary terminal			Spring loaded or “cage clamp” design. Spring washers on stud type
Bus zone CT secondary terminals.			Shorting/disconnecting facility.
Scada terminals			Knife terminal equipped with test 2.3mm sockets
Disconnect/earthing switch characteristics	Part 4 Clause		
Current making rating of earth switches used to apply first earth to feeder, incomer or busbar.	(IEC 62271-102) Clause 6.101		E1
Disconnect bus transfer duty	4.104		Bus transfer duty required
Induced current switching earth switch	4.105		Class B
Rated value of mechanical endurance	4.106		As circuit breaker
Outdoor GIS			
Protection of enclosures against ingress of water	5.13.2		Min IP3XDW
Creepage distance	5.14		In compliance with ENA Technical Specification 41-37
Pollution performance	5.14.101		Class 3
Personal safety clearance	5.107		To include additional 300mm
Air insulated bushings	5.108		In compliance with ENA Technical Specification 41-37

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132kV Switchgear

Summary of Technical Requirements and Ratings	ENA TS 41-37	UNIT	
System Particulars			
Voltage		kV	132
Frequency		Hz	50
Number of Phases			3
Neutral Earthing			Solid
Switchgear General Characteristics	Part 1 Clause		
Number of poles			3
Class – indoor/outdoor	2.1.1		Indoor / Outdoor
Rated voltage	4.1	kV	145
Rated insulation level.	4.2	kV	650
Lightning impulse level			750 (across isolating gap)
Power frequency withstand			275
			315 (across isolator gap)
Rated frequency	4.3	Hz	50
Rated normal current	4.4	A	2000
Rated short-time withstand current	4.5	kA	40
Rated peak withstand current	4.6	kA	100
Rated duration of short circuit	4.7	s	3
Rated supply voltage of opening and closing devices and auxiliary and control circuits:	4.8	V	
Closing and tripping			110
Indication			110
Control			110
Rated supply frequency of closing and auxiliary circuits.	4.9	Hz	DC
GIS Switchgear	Part 2 Clause		
Cable testing	5.202		AC tests either VLF or power frequency will need to be applied at 115kV for 15 minutes.
Circuit breaker characteristics	Part 3 Clause		

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Rated short-circuit breaking current. Equal to rated short-time withstand current.	4.101	kA	40 @ X/R = 14
Transient recovery voltage related to the rated short-circuit breaking current.	4.102		In compliance with ENA Technical Specification 41-37
Rated short-circuit making current	4.103	kA	In compliance with ENA Technical Specification 41-37
Rated operating sequence.	4.104		0 – 0.3 s – CO – 3min - CO
Rated capacitive switching currents : Rated cable-charging current.	4.107	A	Class 2
Rated DC time constant.	4.109	ms	Manufacturer to state (see specified fault breaking current)
Classification of mechanical operations M1 or M2	4.110		Class M1
Additional key interlock			Provision for 1 additional key interlock.
CT secondary terminal			Spring loaded or “cage clamp” design. Spring washers on stud type
Bus zone CT secondary terminals.			Shorting/disconnecting facility.
Scada terminals			Knife terminal equipped with test 2.3mm sockets
Disconnecter/earthing switch characteristics	Part 4 Clause		
Current making rating of earth switches used to apply first earth to feeder, incomer or busbar.	(IEC 62271-102) Clause 6.101		E1
Disconnecter bus transfer duty	4.104		Bus transfer duty required
Induced current switching earth switch	4.105		Class B
Rated value of mechanical endurance	4.106		As circuit breaker
Outdoor GIS			
Protection of enclosures against ingress of water	5.13.2		Min IP3XDW
Creepage distance	5.14		In compliance with ENA Technical Specification 41-37
Pollution performance	5.14.101		In compliance with ENA Technical Specification 41-37
Personal safety clearance	5.107		In compliance with ENA Technical Specification 41-37
Air insulated bushings	5.108		In compliance with ENA Technical Specification 41-37
Bushing terminals	5.109		In compliance with ENA Technical Specification 41-37

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APPENDIX-2: Declaration of Technical Performance to be completed by the manufacturer

MANUFACTURER			
SWITCHGEAR REFERENCE			
	ENATS 41-37 (Part 1 and 2) Reference	UNIT (or compliance)	
General Characteristics	Part 1 Clause		
General	1	Y/N	
Normal service conditions	2.1	Y/N	
Indoor switchgear and controlgear	2.1.1	Y/N	
Outdoor switchgear and controlgear	2.1.2	Y/N	
Number of poles			
Class – indoor/outdoor	2.1.1		
Rated voltage	4.1	kV	
<u>Rated insulation level.</u>	4.2		
Lightning impulse level		kV	
(Across isolating gap)		kV	
Power frequency withstand		kV	
(Across isolating gap)		kV	
Rated frequency	4.3	Hz	
Rated normal current	4.4	A	
Rated short-time withstand current	4.5	kA	
Rated peak withstand current	4.6	kA	
Rated duration of short circuit	4.7	s	
Rated supply voltage of opening and closing devices and auxiliary and control circuits:	4.8		
Closing and tripping		V	
Indication		V	
Control			

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		V	
Rated supply frequency of closing and auxiliary circuits.	4.9	Hz/ DC	
	Part 1 and 2 Clause		
Gas filling valves self sealing?	5.2.104	Y/N	
Gas service valves lockable with ENA standard padlock?	5.2.104.1 5.2.104.2	Y/N	
Identification of pipe-work to BS 1710?	5.2.105	Y/N	
Switchgear able to accommodate recycled SF6 to IEC 60480?	5.2.106	Y/N	
Compartments labelled to indicate pressure and mass of SF6	5.2.107	Y/N	
Earthing of switchgear and control gear generally to IEC 60694	5.3	Y/N	
If outdoor switchgear does switchgear earthing comply with BS 7354	5.3.201 Pt 1	Y/N	
Earthing facilities provided for: Circuit Circuit breaker Busbar	5.3.201 Pt 2	Y/N Y/N Y/N	
Earth-terminals provided for power cable sheath wires	5.3.202 Pt 1	Y/N	
Additional-earth facilities provided provided?	5.3.204	Y/N	
Are provisions made to accommodate SCADA requirements?	5.4.1.102	Y/N	
Does the switchgear contain any auxiliary supplies above 125DC or RMS AC	5.4.4.5.1.101	Y/N	
Do any micro-switches used comply with IEC 60947-1	5.4.4.5.1.103	Y/N	
Auxiliary wiring conductor size: CT secondary wiring. Control secondary wiring	5.4.4.5.1.105	mm mm	
CT secondary wiring terminal blocks		Type	
CT bus wire terminal blocks		Type	
SCADA wiring terminal blocks		Type	
Control wiring terminal blocks		Type	
Low and high pressure interlocking and monitoring devices comply with 5.9 of IEC 60694	5.9	Y/N	

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Gas monitoring: Remote alarms	5.9.101		
Circuit breaker: Stage 1: Pressure / function Stage 2: Pressure / function		barg barg	
Busbar: Stage 1: Pressure / function Stage 2: Pressure / function		barg barg	
Feeder: Stage 1: Pressure / function Stage 2: Pressure / function		barg barg	
Hydraulic/pneumatic pressure monitoring: Alarm. Pressure/function No of volt free contacts	5.9.102		
Temporary blocking of gas density alarms	5.9.102	Y/N	
Dual pressure systems: Compliance with requirement.	5.9.103	Y/N	
Replacement of gas sensors: Compliance with requirement.	5.9.104	Y/N	
Name plates	5.10	Y/N	
Interlocking and padlocking	5.11	Y/N	
Circuit breaker, disconnecter interlocking	5.11.101	Y/N	
Earth interlock	5.11.102	Y/N	
Gas and vacuum tightness	5.15 5.15.3	Y/N	
Ergonomics and access	5.101	Y/N	
Internal fault	5.102	Y/N	
External effect	5.102.2	Y/N	
Enclosures	5.103	Y/N	
Mechanism cabinets for outdoor switchgear	5.104 (Pt1)	Y/N	
Partitions	5.104 (Pt 2)	Y/N	
Limitation of pressure rise in case of internal fault	5.105.2	Y/N	
Surface preparation and coating	5.106	Y/N	
Expected operating life	5.111	Years	
Pressure relief	5.112	Y/N	
Test facilities	5.201	Y/N	
Test bushings	5.201.1	Y/N	

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Cable testing	5.202	Y/N	
Cable testing: Can test bushings accommodate AC cable testing requirement? (see appendix 1 of this document)		Y/N	
Test bushings – Dc testing	5.202.1	Y/N	
Cable DC testing	5.202.2	Y/N	
Primary injection testing of CTs	5.203	Y/N	
Outage constraints	5.204	Y/N	
Type tests	6	Y/N	
Routine tests	7	Y/N	
Circuit breaker characteristics	Part Clause 3		
General	1	Y/N	
Rated short-circuit breaking current. Equal to rated short-time withstand current.	4.101	kA	
Transient recovery voltage related to the rated short-circuit breaking current.	4.102	Y/N	
Rated short-circuit making current	4.103	kA	
Rated operating sequence.	4.104		
Characteristics for short line faults	4.105	kA	
Out of phase making and breaking.	4.106	kA	
Rated capacitive switching currents : Rated cable-charging current.	4.107	A	
Rated small inductive breaking current.	4.108	A	
Rated DC time constant.	4.109	Ms	
Classification of mechanical operations M1 or M2	4.110		
Trip coil supervision	5.4.101	Y/N	
Trip coil isolation	5.4 102	Y/N	
Forced opening	5.4.103	Y/N	
Anti pumping	5.5.101	Y/N	
Operating systems interlock	5.5.102	Y/N	
Circuit breakers with independent drive mechanisms	5.5.103	Y/N	
Non simultaneous pole position	5.5.104	Y/N	
Stored energy closing	5.6	Y/N	

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Three pole operation	5.6.101	Y/N	
Tripping mechanism	5.6.102	Y/N	
Local manual tripping	5.8.201	Y/N	
Slow closing and opening mechanism?	5.8.202	Y/N	
Interlocked with CB isolation.		Y/N	
Operations counter	5.8.203	Y/N	
Nameplates: State DC time constant?	5.10	Y/N	
Trip and closing coil isolation: labelled.	5.10.101	Y/N	
Mechanical key interlocking	5.11.101	Y/N	
Circuit identification	5.202	Y/N	
Noise	5.203	Y/N	
CT labelling	5.204	Y/N	
Type tests	6	Y/N	
Short circuit making current	6.104.2	Y/N	
Test duty T30	6.104.5.4	Y/N	
Out of phase conditions	6.110	Y/N	
Capacitive current switching tests	6.111	Y/N	
Sound pressure measurement	6.201	Y/N	
Breaking tests with asymmetrical current	6.202	Y/N	
Routine tests	7	Y/N	
Disconnecter/Earthing switch characteristics	Part 4 Clause		
Current making rating of earth switches used to apply first earth to feeder, incomer or busbar.	(IEC 62271-102) Clause 6.101	Class	
Disconnecter bus transfer duty	4.104	Rating	
Induced current switching earth switch	4.105	Class	
Rated value of mechanical endurance	4.106	Class	
Outdoor GIS switchgear			
Degree of protection for enclosures for outdoor equipment.	5.13.2	Class	
Creepage distance of external bushings	5.14	mm	

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Pollution performance of external bushings	5.14.101	Class	
Personal safety clearance (Additional 300mm)	5.107	Y/N	
Air insulated bushings	5.108	Y/N	
Bushing gauges (Where liquid filled)	5.109	Y/N	
Bushing terminals	5.110	Type	

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TYPE TEST CONFORMANCE DECLARATION 2.1

Manufacturer:..... **Ratings**.....

Product Reference:.....

Name:

Signature:

Date:.....

Instructions for completion:

- Complete a separate table for each variant and rating
 - Manufacturer to complete columns 5 to 10
 - When test report also covers another rating insert 'See ???A unit' in the Remarks column
- Tests not requested may be shown as 'Additional tests' at the bottom of the table

*See bottom of table for conformance declaration codes

** I = Independent; M= Manufacturer; ENA= Energy Networks Association

	Test Requirement	Specification and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA**	Remarks
1	Dielectric Circuit breaker-disconnector and earthing switches as appropriate. Partial discharge	IEC 60694.subclause6.2, IEC 62271-203. Clause 6.2, IEC 62271-100. Subclause6.2. Subclause 7.101 IEC62271-103 ENA TS 41-18								
2	Radio interference test	Subclause 6.3 IEC 62271-203								
3	Measurement of the resistance of main circuit -	Subclause 6.4 IEC 62271-203								
4	Temperature Rise	Subclause 6.5, IEC 62271-203.								
5	Short-time withstand current and peak withstand current tests -	Subclause 6.6, IEC 62271-203.								
6	Demonstration of arcing times	Subclause 6.102.10, IEC 62271-100								
7	Three phase tests	Subclause 6.102.10.1, IEC 62271-100								

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	Test Requirement	Specification and Standards	Rated value	Test req'd Y or N	Conformance	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA**	Remarks
8	Test duties T10,T30,T60,T100s(b).OP 1 and OP2	Subclause 6.102.10.1.1, IEC 62271-100								
9	Test duty T1 00a	Subclause 6.102.1.2, IEC 62271-100								
10	Tests on earthing circuits	Subclause 6.6.102 IEC 62271-203.								
11	Verification of protection	Subclause 6.7, IEC 62271-203.								
15	EMC tests	IEC 60694.subclause 6.9								
16	Gas tightness test	IEC 60694 subclause 6.8 & IEC 62271-203 subclause 6.8								
17	Electromagnetic compatibility	IEC 60694 subclause 6.9 & IEC 600271-203 subclause 6.9								
18	Mechanical and environmental tests – circuit breaker.	IEC 62271-203 Subclause 6.102, IEC 62271-100. Subclause 6.101								
19	Mechanical and environmental tests –disconnecter and earthing switches.	IEC 62271-203 subclause 6.102, IEC 62271-102. subclause 6.102								
20	Proof tests for enclosures	IEC 62271-203. subclause 6.103								
21	Pressure test on partitions	IEC 62271-203. subclause 6.104								
20	Internal fault	IEC 62271-203. subclause 6.105								
21	Insulator tests	IEC 62271-203. subclause 6.106								
22	Additional tests on auxiliary and control circuits	Subclause 6.10 IEC 60694								
23	Process Control	ISO 9000 ERG79 Parts 1 & 2a								

* Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product
 Cs1 = the test conforms fully with the requirements of this clause
 Cs2 = the test conforms partially with the requirements of this clause
 Cs3 = the test
 d Cs4 = Test not performed, but alternative evidence/ technical case offered

Ct1 = Independent witnessed tests
 Ct2 = Not fully independent witnessed tests
 Ct3 = Self verified tests
 es Ct4 = Alternative tests / evidence offered
 Ct5 = Manufacturer has underwritten that the product meets the functional And performance requirements without further testing.
 Ct6 = Not tested

CAUTION! - This document may be out of date if printed

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Appendix 3: Addendum To Supplier Requirements

Project specific requirements will be provided by Primary Engineering Projects for inclusion in this appendix.